ENEL 585 – Lab 3

AC-DC Conversion: Single-Phase and Three-Phase Diode Bridge Rectifiers — DC-Side Inductor, Commutation and Power Factor

Group Information

Group #: _____

Student name(s): _____

Purpose

To operate the diode bridge of the SEMITEACH box as a single-phase and as a three-phase rectifier, supplied at 60 Hz from the station's transformer box, and to measure how a dc-side inductor changes the dc current, the shape of the source current and hence the power factor. You will compare the measured waveforms, average output voltages, harmonics and power factor with the idealised results from class, test whether the assumptions behind them hold with the station's components, and measure the commutation overlap caused by the transformer's own leakage inductance.

Marking

Total Marks for Pre-Lab Quiz = __ /10 (Weight = 2%)
Total Marks for Lab Report = __ /10 (Weight = 2%)
Total Marks for Lab Work = __ /10 (Weight = 2%)

The pre-lab itself is not collected or graded. Instead, a **10-minute individual quiz** is written at the start of the lab session, based directly on the pre-lab questions below.

Associated Documents

- *Bench Instruments Guide* — front-panel procedures for the oscilloscope and probes (§4).
- *Lab Resources* — photographs and ratings of the transformer box, its Acuvim-L power meter and the SEMITEACH box.

Introduction

Source. The transformer box steps the station's 208 V three-phase supply down to V_{LL} = 36.7 V rms line-to-line (21.2 V line-to-neutral) at 60 Hz, through a wye-wye transformer rated 1.2 kVA. Its output terminals are A, B, C, N (neutral) and G (ground). It has two breakers, one on the high-voltage (208 V) side and one on the low-voltage side, and an Acuvim-L power meter on its front panel. The transformer's leakage inductance appears in series with each line: it is the **commutation inductance L_c** of this lab. You do not add one, and its value is not given — Experiment 4 measures it.

Rectifier. The SEMITEACH box contains a three-phase diode bridge (SKD 51/14) with its own *Rectifier input* terminals and *DC rectifier output* terminals (red +, blue −). The bridge output is not connected inside the box to the inverter's DC link. Part A uses two of the three inputs as a single-phase bridge and leaves the third unconnected; Part B uses all three.

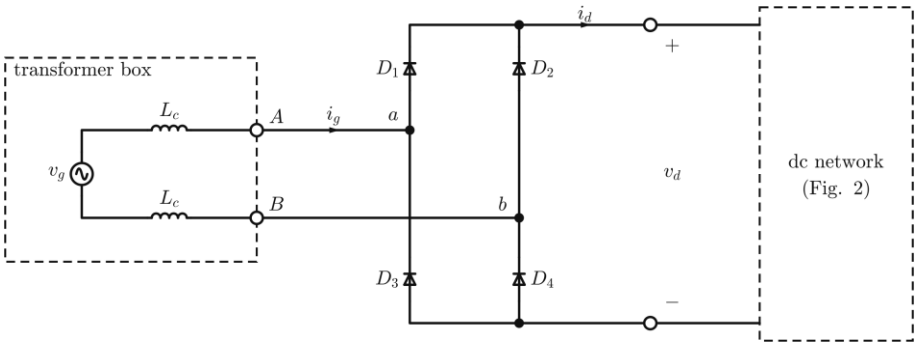


Fig. 1 – Part A: single-phase bridge fed line-to-line from transformer terminals A and B. v_g and L_c represent the transformer: L_c is the leakage inductance of one line. D_1–D_4 are four of the SKD 51/14 diodes; the third rectifier input is left unconnected. The dc network is one of Fig. 2.

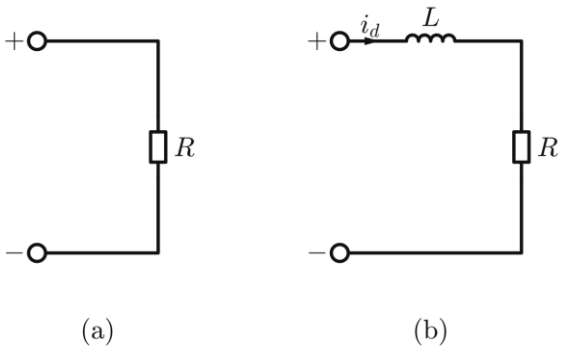


Fig. 2 – Networks connected to the bridge's dc terminals: (a) R; (b) L–R. L = 5 mH (Hammond 195G20), R = resistor load bank.

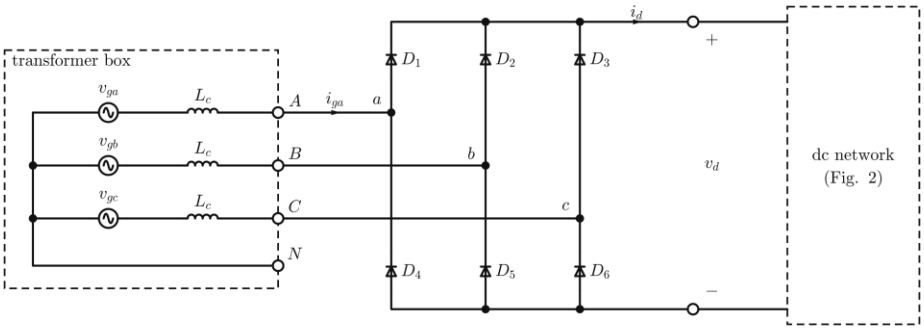


Fig. 3 – Part B: three-phase bridge D_1–D_6 fed from transformer terminals A, B and C. L_c is the leakage inductance of one line. The neutral N is used only as the reference for measuring v_{aN}.

Measured quantities. In Part A the source voltage is measured across the bridge input, v_{ab} (a to b in Fig. 1), and i_g is the current in line a. In Part B the phase voltage v_{aN} is measured from bridge input a to the transformer neutral, and i_{ga} is the current in line a. In both parts v_d is the bridge output voltage and i_d the bridge output current; $\langle v_d \rangle$ and I_d are their averages, as in class. Because L_c sits inside the transformer, v_{ab} and v_{aN} are the transformer's terminal voltages, not the ideal source v_g.

Power factor terms, as in class: PF = P/($V_{rms}I_{rms}$) at the rectifier input (k_p in class), and THD = $\sqrt{(I_{rms}^2 - I_1^2)}/I_1$, the total harmonic distortion of the current relative to its rms fundamental I_1. With a constant i_d and no overlap, the ideal values are THD = 48 % and PF = 0.90 for the single-phase bridge (from class) and THD = 31 % and PF = 0.955 for the three-phase bridge.

Pre-Lab

Work these before the session; the pre-lab quiz at the start of the lab is drawn from them. Keep your calculated values: your report compares the measurements with them.

Parameters (all questions): V_{LL} = 36.7 V rms at f = 60 Hz. The single-phase bridge is fed line-to-line, so v_g has a peak of $\hat{V}_g = \sqrt{2} \times 36.7 = 51.9$ V. In the three-phase bridge each phase voltage v_{ga}, v_{gb}, v_{gc} has a peak of $\sqrt{2} \times 21.2 = 30.0$ V. R = 12.5 Ω, L = 5 mH. Assume ideal diodes.

1. **Single-phase bridge, network (a), R = 12.5 Ω.** Calculate $\langle v_d \rangle$ and I_d.
2. **Single-phase bridge, network (b), L–R.** The ripple in v_d is at 120 Hz. Compare the inductor's reactance at 120 Hz, X = 2π × 120 × L, with R. Is L = 5 mH large enough to hold i_d nearly constant?
3. **Three-phase bridge, network (b), R = 12.5 Ω, treating i_d as constant.** Calculate $\langle v_d \rangle$ and I_d with L_c = 0. Then assume each line has L_c = 100 µH, a typical value (Experiment 4 measures the real one). Using the results from class, find the overlap angle u, in degrees and in µs.
4. **Power factor.** From class, the single-phase bridge with constant i_d has PF = 0.90, although i_g is in phase with v_g. Why is it not 1?

Safety Notes

- The 208 V station supply from the ABB box plugs in only at the **back** of the transformer box, into the inputs labelled HIGH VOLTAGE. Never plug it into the front-panel terminals A, B, C, N, G: they are the low-voltage side of the transformer, rated 36.7 V. 208 V there drives the transformer far into saturation, draws a very large current and can put a dangerous voltage on the high-voltage side. The high-voltage side stays live between experiments, so do not touch the 208 V cable or the HIGH VOLTAGE inputs while the ABB supply is on.
- Before the 208 V supply from the ABB box is switched on, check that both transformer box breakers, high-voltage and low-voltage side, are off.
- The voltages in this lab are modest (36.7 V rms line-to-line, 52 V peak), but the transformer can drive a large current into a short circuit. Make every change — wiring, moving a probe, changing R, inserting or removing the inductor — with the transformer box switched off; never open the dc circuit or a load-bank switch while current flows.
- Measure every voltage with the differential probes. Do not connect an oscilloscope probe ground or any other earthed lead anywhere in the circuit. The transformer neutral N is used only as a measurement reference in Part B; G is not used.
- Nothing is connected to the SEMITEACH box except its rectifier terminals, and nothing but the networks of Fig. 2 is connected to the DC rectifier outputs.
- Obtain TA sign-off before first switching the transformer on, and again after re-wiring for Part B.

Equipment

Component	Purpose	Qty
Oscilloscope (Keysight DSOX2004A)	Display probed signals	1

Component	Purpose	Qty
Keysight 1146B clamp-on current probe	Measure i_g (or i_{ga}) and i_d for the oscilloscope	2
Keysight N2791A differential voltage probe	Measure v_{ab} (or v_{aN}) and v_d	2
Transformer box	60 Hz source: 36.7 V line-to-line, terminals A, B, C, N, G, high- and low-voltage breakers; Acuvim-L meter for rms values, power, THD and harmonics	1
Inverter box (SEMITEACH)	Diode bridge only: Rectifier input and DC rectifier output terminals	1
5 mH inductor	L in Fig. 2	1
Resistor load bank	R in Fig. 2: 20 × 250 Ω in parallel, one switch each — all 20 = 12.5 Ω	1
Banana and BNC cables	Power circuit and probes	—

Instrument Settings for Lab 3

The TA checks these settings at sign-off:

- **Oscilloscope:** set it up by hand, not with [Auto Scale], starting from Table 1. Read averages and rms values with [Meas] over a whole number of 60 Hz periods (Guide §4).
- **Acuvim-L meter** (transformer box): the meter is set up in advance; do not change its settings. [V/A] steps through the voltages and line currents and [P] through the powers; holding [H] and [V/A] together for about 1 s and selecting *Harmonic* gives the current THD and the individual harmonics, both relative to the fundamental as in class. Power factor: in Part B, read the total PF from the meter's PF page. In Part A, calculate PF = $P/(V_{AB}I_A)$ from the meter's total P, V_{AB} and I_A; the meter's PF readings do not apply there, because it refers each phase to the line-to-neutral voltage while the single-phase bridge is fed line-to-line. Let readings settle for a few seconds.
- **Accuracy:** the oscilloscope's dc accuracy is about ±3 % of full scale (8 divisions): ±2.4 V at 10 V/div, ±0.24 A at 1 A/div, plus a few percent for each probe. The Acuvim-L is specified at ±0.5 % for V, I and P and ±1 % for harmonics (its manual), and its current transformers add their own error at these small currents. Use these figures when you judge whether a difference from a prediction is significant.

Table 1: Suggested starting oscilloscope settings — adjust as needed

View	Time/div	Ch 1, Ch 2 (voltages)	Ch 3, Ch 4 (currents)	Trigger
Exp. 1–2: single phase	5 ms	v_{ab} 20 V/div, 0 V at centre; v_d 10 V/div, 0 V three divisions below centre	i_g 2 A/div, 0 A at centre; i_d 1 A/div, 0 A three divisions below centre	v_{ab}, rising, 0 V, Auto
Exp. 3–4: three phase	5 ms	v_{aN} 10 V/div, 0 V at centre; v_d 10 V/div, 0 V three divisions below centre	i_{ga} 2 A/div, 0 A at centre; i_d 1 A/div, 0 A three divisions below centre; zoom in on i_d to read its ripple	v_{aN}, rising, 0 V, Auto

View	Time/div	Ch 1, Ch 2 (voltages)	Ch 3, Ch 4 (currents)	Trigger
Exp. 4: commutation	100 µs, then finer	as above	i_{ga} and i_d 1 A/div, then 0.5 A/div	i_{ga}, rising, near $I_d/2$, Normal

Setup — Single-Phase Bridge (Part A)

The TA checks the wiring before the transformer is switched on.

1. Check that both transformer box breakers are off and that the 208 V cable from the ABB box is plugged into the HIGH VOLTAGE inputs at the back of the box, not the front panel. Nothing is connected to the SEMITEACH box except its rectifier terminals: no +15 V supply, no BNC cables, nothing on the DC-link or inverter terminals.
2. Power circuit of Fig. 1 with banana cables: transformer terminal A → first *Rectifier input* (node a); terminal B → second *Rectifier input* (node b). The third rectifier input and the transformer's C, N and G terminals stay unconnected.
3. DC side, network (a) of Fig. 2: *DC rectifier output* + (red) → resistor load bank → − (blue). Close **all 20 load-bank switches** (12.5 Ω).
4. Voltage probes (10:1): channel 1 across the rectifier inputs a (+) and b (−) for v_{ab}; channel 2 across the DC rectifier outputs, + to +, for v_d. Current probes (100 mV/A, zeroed): channel 3 on the line-a wire, arrow toward the rectifier; channel 4 on the wire from the red DC output, arrow away from the rectifier.

STOP: TA to verify the circuit and instrument settings before the transformer is switched on.

Startup and shutdown. To energize the transformer box, close the high-voltage breaker first, then the low-voltage breaker. To de-energize it, open the low-voltage breaker first, then the high-voltage breaker. Between experiments, "switch the transformer on/off" means the low-voltage breaker only; the high-voltage breaker stays closed. Every experiment: switch the transformer on, take the measurements, and switch it off.

Table 2: The four experiments, in the order you run them

Exp.	Bridge	Dc network (Fig. 2)	R [Ω]
1	Single phase (A–B)	(a) R	12.5 (all 20 switches)
2	Single phase (A–B)	(b) L–R, L = 5 mH	12.5
3	Three phase	(a) R	12.5
4	Three phase	(b) L–R, L = 5 mH	12.5

Part A — Single-Phase Bridge

Read the questions before you start: you cannot repeat the measurements after the session. For each experiment, capture v_{ab}, v_d, i_g and i_d over at least two periods. From the oscilloscope, record at least the averages $\langle v_d \rangle$ and I_d and the minimum and maximum of i_d; from the meter, V_{AB}, I_A, the total P and the THD of i_g. In Experiment 2, also record the harmonics of i_g up to the 13th.

Experiment 1

Questions:

- Compare $\langle v_d \rangle$ and I_d with pre-lab question 1, and the PF with what you expect for a resistive load.
- The load is a resistor, yet the meter shows some distortion in i_g. Where does it come from?

Experiment 2

With the transformer off, insert the inductor between the red DC output and the load bank (Fig. 2(b)). Besides the usual readings, use the oscilloscope's FFT (Guide §4) to measure the 120 Hz component of v_d.

Questions:

- Compare i_d and i_g, and the THD and PF, with the constant-i_d results from class (Introduction) and with Experiment 1. Does what you see agree with your answer to pre-lab question 2?
- How large would L need to be to keep the peak-to-peak ripple of the load voltage v_R within 10 % of its average at R = 12.5 Ω? Base your estimate on the 120 Hz component of v_d you measured. Would such an inductor be practical?

Part B — Three-Phase Bridge

Switch the transformer off. Take the inductor out and restore network (a), with all 20 load-bank switches closed. Add line C: transformer terminal C → third *Rectifier input* (node c). Move the channel 1 probe to measure v_{aN}: + on rectifier input a, − on the transformer's N terminal. The current probes stay where they are: channel 3 now measures i_{ga}. For each experiment, capture v_{aN}, v_d, i_{ga} and i_d, and record what you recorded in Part A, plus the minimum and maximum of v_d, all three line voltages and currents, and the meter's total PF. In Experiment 4, also record the harmonics of i_{ga} up to the 13th.

STOP: TA to verify the Part B wiring before the transformer is switched on.

Experiment 3

Questions:

- Compare $\langle v_d \rangle$, I_d and the minimum and maximum of v_d with what you expect from class (pre-lab question 3, $L_c = 0$).
- Explain the shape of i_{ga}, marking on your capture which diodes conduct in each interval.

Experiment 4

With the transformer off, insert the inductor (Fig. 2(b)) and take the same measurements as in Experiment 3. Then zoom in on one commutation: capture the rise of i_{ga} together with i_d on a time base fine enough to show the whole rise, and measure the overlap u with the cursors: from where i_{ga} leaves zero to where it **first** reaches the value of i_d at that instant (put a horizontal cursor on the i_d trace). i_{ga} rises slowly at first, so use the most sensitive current scale that keeps the whole rise on screen. Ignore the brief overshoot and snap just after the rise: that is reverse recovery of the outgoing diode, not overlap.

Questions:

- Compare I_d with pre-lab question 3, and the THD and PF with the ideal three-phase values (Introduction).
- Compare the harmonics of i_{ga} with those of i_g in Experiment 2. Which orders appear in each, and why?

- From your capture, estimate the overlap angle u and the commutation inductance L_c of one line, and compare u with pre-lab question 3.

Comparison

Questions:

- Tabulate, for Experiments 1–4: $\langle v_d \rangle$, the peak-to-peak ripple of i_d as a fraction of I_d, the THD and the PF. Rank the four cases by PF and explain the ranking from the shape of the source current.
- Compare the change in i_d and in the source current from Experiment 3 to Experiment 4 with the change from Experiment 1 to Experiment 2, and explain the difference between the single-phase and three-phase bridges.
- If you were to supply a 1 kW resistive dc load from this transformer with a diode rectifier, which bridge would you choose, and would you fit a dc-side inductor? State what you would trade off.

Shutdown

1. Switch the transformer box off: open the low-voltage breaker, then the high-voltage breaker.
2. Turn off the current probes, disconnect your wiring, and return the lab equipment as instructed by the TA.

Lab Report

Submit one report per group on D2L by 10:00 AM one week after your lab session. The report is marked out of 10 using the rubric below, rather than experiment by experiment. Include the required captures and measurements, answer the questions directly, and use your results to support your explanations. Do not narrate the procedure.

Table 3: Lab report rubric (10 marks)

Criterion	Marks	Full-marks standard
Experimental evidence	3	Required captures and measurements are present and usable. Channels, scales, zero levels, probe factors, operating conditions and units are clear where applicable. Relevant measurement limitations are identified.
Analysis and comparison	3	Calculations are correct. Measurements are compared quantitatively with applicable theoretical predictions, with assumptions stated. Relevant trends across operating conditions are identified.
Physical interpretation	4	Answers explain the observed behaviour using switching states, current paths and circuit principles. Significant discrepancies are discussed using the available evidence, with uncertainty acknowledged.

Table 4: Quality scale, applied separately to each criterion

Level	Share of marks	Standard
Excellent	90–100%	Complete and technically correct; no substantive weaknesses.
Good	75–89%	Substantially complete and correct; minor weaknesses do not undermine the results or conclusions.
Adequate	60–74%	Demonstrates the essential requirements, but omissions or errors limit the analysis, interpretation or usefulness of the evidence.
Marginal	50–59%	Provides some relevant evidence or understanding, but substantial omissions or errors undermine the results or conclusions.

Level	Share of marks	Standard
Inadequate	below 50%	Provides little usable evidence of meeting the criterion, or contains fundamental errors. No relevant work earns zero.